Partial Differential Equations 3 (Numerical)
Lecture 1 - Academic Year 2025/2026

Prof Halim Kusumaatmaja
Email: halim.kusumaatmaja@ed.ac.uk

Semester 2, Edinburgh Engineering

Aims and Objectives

1. To explain how 2nd order linear PDEs can be categorised.
2. To discuss finite difference concept and its basic schemes.

General Form of Partial Differential Equations

When solving partial differential equations, we are generally interested in finding a solution to the function $u(x, y, z, t)$ that depends on the positions in three dimensions (x, y, z) and time t. In general a partial differential can be written as:

$$F(x, y, z, t; u, u_x, u_y, u_z, u_{xx}, u_{yy}, u_{zz}, u_{xy}, u_{xz}, u_{yz}, \cdots) = 0. \tag{1}$$

In the above equation, we have used the subscript notation for the partial derivatives such that

$$\begin{gathered} u_x = \frac{\partial u}{\partial x}; \quad u_y = \frac{\partial u}{\partial y}; \quad u_z = \frac{\partial u}{\partial z}; \\ u_{xx} = \frac{\partial^2 u}{\partial x \partial x}; \quad u_{yy} = \frac{\partial^2 u}{\partial y \partial y}; \quad u_{zz} = \frac{\partial^2 u}{\partial z \partial z}; \\ u_{xy} = \frac{\partial^2 u}{\partial x \partial y}; \quad u_{xz} = \frac{\partial^2 u}{\partial x \partial z}; \quad u_{yz} = \frac{\partial^2 u}{\partial y \partial z}. \end{gathered}$$

A PDE is said to be linear if the general equation in Eq. (1) is linear in u and its derivatives. Linear PDEs are much easier to solve than non-linear PDEs.

Exercise 1: Which PDE is linear and non-linear (please find the equations online or in textbooks): (a) Navier-Stokes equation, (b) Cahn-Hilliard equation, and (c) heat equation.

We can define the order of the PDE by looking at the highest partial derivative appearing in Eq. (1). For example, first order linear PDEs will have terms dependent on u_t, u_x, or both, but not their products; while second order linear PDEs will have terms dependent on u_{xx}, u_{xy}, and so on. In this course, we will focus on 2nd order linear PDEs. For this class of PDEs, the general PDE equation can be reorganised to have the following form:

$$\sum_{i,j=1}^{n} a_{ij} u_{ij} + \sum_{i=1}^{n} b_i u_i + cu = d. \tag{2}$$

To avoid future confusion, we will interchangeably use subscripts that are explicitly written as (x, y, z, t), or we may write them in terms of $(i, j, \cdots) = (1, 2, \cdots)$. The former makes the equations more descriptive and are common in engineering problems, while the latter is handy when trying to write the equations in a more general way.

Another terminology that you will often see in literature: If $d = 0$ we call the PDE homogeneous, while the PDE is inhomogeneous when $d \neq 0$.

Elliptic, Parabolic and Hyperbolic Equations

Since the partial derivatives are symmetric, i.e., $u_{ij} = u_{ji}$, the first term in Eq. (2) can be re-arranged in the form of a real, symmetric, co-efficient matrix $A = [a_{ij}]$. Such matrix will have n real eigenvalues, where n is the number of independent dimensions. For instance, $n = 3$ if we have relevant partial derivatives in (x, y, z) coordinates. The number of zero eigenvalues Z and positive eigenvalues P determine the nature of the PDEs and in turn how we may solve them. We are interested in the following cases:

- **elliptic** if $(Z = 0$ and $P = 0)$ or $(Z = 0$ and $P = n)$;
- **parabolic** if $\det A = 0$;
- **hyperbolic** if $(Z = 0$ and $P = 1)$ or $(Z = 0$ and $P = n - 1)$.

Alternatively, if we have a two-dimensional PDE,

$$a_{11}u_{11} + 2a_{12}u_{12} + a_{22}u_{22} + \cdots = 0, \tag{3}$$

we can define the PDE to be elliptic if $a_{12}^2 - a_{11}a_{22} < 0$, parabolic if $a_{12}^2 - a_{11}a_{22} = 0$ and hyperbolic if $a_{12}^2 - a_{11}a_{22} > 0$.

Let us now do an example. Consider the Laplace equation:

$$\nabla^2 u = u_{xx} + u_{yy} + u_{zz} = 0. \tag{4}$$

The matrix A is therefore given by:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{5}$$

Recall that to get the eigenvalues, we usually write the equation as $A = \lambda I$, where I is the identity matrix, and the eigenvalues are the roots of $\det(A - \lambda I) = 0$. In the case of the Laplace equation we find three positive, repeated, eigenvalues $\lambda = 1$. Following our classification we have $P = 3$ and $Z = 0$, which indicates that the Laplace equation is an elliptic PDE.

Laplace Equation is prominent in many areas of science and engineering. Examples include:

- electrostatics: $\nabla^2 V = 0$, where V is the electric potential.
- 2-D irrotational flow: $\nabla^2 \phi = 0$, where the fluid velocity is defined as $\mathbf{v} = -\nabla\phi$.
- steady state heat equation: $\nabla^2 T = 0$, where T is the temperature.
- and many more.

Another important elliptic PDE that you will see often is the Poisson equation, which is essentially a Laplace equation with a source term. Elliptic PDEs tend to be steady state equations without any time dependence.

Exercise 2: Can you determine whether the following PDEs are elliptic, parabolic or hyperbolic:

$$\text{(i)} \qquad \frac{\partial T}{\partial t} - \kappa\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) = 0.$$

$$\text{(ii)} \qquad \frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2} = 0.$$

Finite Difference Scheme

Thus far, we have discussed the PDEs in their analytical forms. This part of the course is on how to solve them numerically. There are many ways to do this. In our course, we will explore a class of approaches under the umbrella of finite difference methods. Conceptually this is the most basic way of approaching PDEs numerically, and they are often involved even when we are employing other numerical approaches.

Consider now the case where space and time are discrete, separated by Δx, Δy, Δz, and Δt. In principle different ways of discretisation are possible, but in this course we will usually discretise space and time uniformly (the same everywhere) and use square grid for 2D problems, cubic grid for 3D problems, etc.

With this in mind, we can consider how we may write down the partial derivatives. Consider the simplest derivative of u_x. We may write this in the so-called forward difference scheme:

$$\text{Forward} : u_x = \frac{\partial u}{\partial x} = \frac{u(x+\Delta x) - u(x)}{\Delta x}. \tag{6}$$

Alternatively we may have written this in the so-called backward and central difference schemes:

$$\text{Backward} : u_x = \frac{\partial u}{\partial x} = \frac{u(x) - u(x-\Delta x)}{\Delta x}. \tag{7}$$

$$\text{Central} : u_x = \frac{\partial u}{\partial x} = \frac{u(x+\Delta x) - u(x-\Delta x)}{2\Delta x}. \tag{8}$$

Ideally we want to use $\Delta x \to 0$ and in this case all schemes will give accurate results. However, in practice, using $\Delta x \to 0$ leads to very long and expensive calculations. In fact, due to numerical precision on our computers, employing too small Δx can lead to poor results. Hence, we want to use Δx that is small enough but not too small. If we can have a better scheme, we can use larger spacing / grid size / time step, leading to cheaper and more effective calculations.

Fig. 1 illustrates an example function $f(x)$, and we carry out forward, backward, and central difference schemes to estimate the gradient/slope of the curve at the point indicated by the black circle. We use two different values of Δx in the top and bottom panels. The analytical expectation is also shown. For the larger Δx we can clearly see that the central difference scheme is quite close to the analytical result, while the forward and backward difference schemes are inaccurate. For smaller Δx all schemes give reasonable estimates for the gradient. To help you develop intuition, visit the link below for an interactive tool where you can vary Δx and see the results for u_x for the three schemes discussed above. As we shall see in the next section, by employing Taylor series, we can rationalise that the central difference is more accurate because it is a second order accurate scheme, while the forward and backward difference schemes are only first order accurate. Generally we want numerical schemes that are second order accurate.

Taylor Series

To rationalise the order of accuracy of finite difference schemes, we can take advantage of Taylor series. Recall that we can write:

$$\begin{aligned}
u(x+\Delta x) &= u(x) + \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2!}\frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^3}{3!}\frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{4!}\frac{\partial^4 u}{\partial x^4} + \cdots \\
u(x-\Delta x) &= u(x) - \Delta x \frac{\partial u}{\partial x} + \frac{\Delta x^2}{2!}\frac{\partial^2 u}{\partial x^2} - \frac{\Delta x^3}{3!}\frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^4}{4!}\frac{\partial^4 u}{\partial x^4} + \cdots
\end{aligned}$$

Using these equations, our forward, backward and central difference schemes give:

$$\begin{aligned}
\text{Forward} : \quad & \frac{u(x+\Delta x) - u(x)}{\Delta x} = \frac{\partial u}{\partial x} + \frac{\Delta x}{2!}\frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{3!}\frac{\partial^3 u}{\partial x^3} + \frac{\Delta x^3}{4!}\frac{\partial^4 u}{\partial x^4} + \cdots \\
\text{Backward} : \quad & \frac{u(x) - u(x-\Delta x)}{\Delta x} = \frac{\partial u}{\partial x} - \frac{\Delta x}{2!}\frac{\partial^2 u}{\partial x^2} + \frac{\Delta x^2}{3!}\frac{\partial^3 u}{\partial x^3} - \frac{\Delta x^3}{4!}\frac{\partial^4 u}{\partial x^4} + \cdots \\
\text{Central} : \quad & \frac{u(x+\Delta x) - u(x-\Delta x)}{2\Delta x} = \frac{\partial u}{\partial x} + \frac{\Delta x^2}{3!}\frac{\partial^3 u}{\partial x^3} + \cdots
\end{aligned}$$

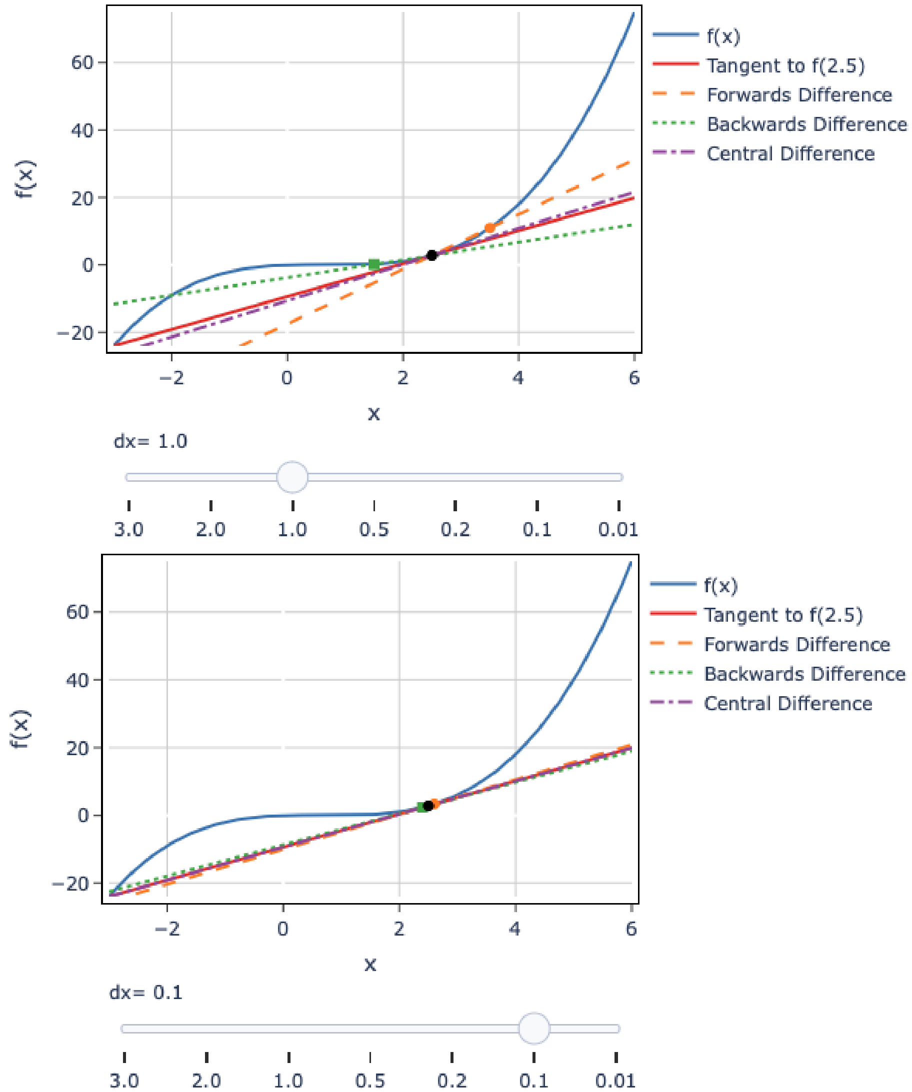

Figure 1: Comparisons of forward, backward and central finite difference schemes for two different values of Δx.

We can observe that both for the forward and backward difference schemes, we get the term we want (i.e., u_x) and the highest order error term is proportional to u_{xx}. Since the error term is one order higher than the term we want, we call this a first order accurate scheme. In contrast, for the central difference scheme, the u_{xx} term cancels and the highest order error term is proportional to u_{xxx}. This is two orders higher, which is why we call it a second order accurate scheme.

Exercise 3: Show that the scheme below for u_{xx} is second order accurate scheme:

$$u_{xx} = \frac{u(x+\Delta x) + u(x-\Delta x) - 2u(x)}{\Delta x^2}. \tag{9}$$

Two and Three Dimensional Partial Derivatives

Since we are interested in second order linear PDEs, in 2D and 3D, we will have mixed partial derivatives. Without any loss of generality, we can just consider the 2D case since in 3D the simplest second order accurate mixed derivatives can be done in the relevant 2D planes, which gets us back to the 2D results. Of course, more elaborate schemes in 3D are available. In general, we can carry out the following Taylor series expansion:

$$
\begin{aligned}
u(x+\Delta x, y+\Delta y) &= u(x,y) + \Delta x\frac{\partial u}{\partial x} + \Delta y\frac{\partial u}{\partial y} \\
&+ \frac{\Delta x^2}{2!}\frac{\partial^2 u}{\partial x^2} + \frac{2\Delta x\Delta y}{2!}\frac{\partial^2 u}{\partial x\partial y} + \frac{\Delta y^2}{2!}\frac{\partial^2 u}{\partial y^2} \\
&+ \frac{\Delta x^3}{3!}\frac{\partial^3 u}{\partial x^3} + \frac{3\Delta x^2\Delta y}{3!}\frac{\partial^3 u}{\partial x^2\partial y} + \frac{3\Delta x\Delta y^2}{3!}\frac{\partial^3 u}{\partial x\partial y^2} + \frac{\Delta y^3}{3!}\frac{\partial^3 u}{\partial y^3} + \cdots
\end{aligned}
$$

Exercise 4: Show that the central difference schemes we have derived in the previous section maintains their second order accuracy for u_x, u_y, u_{xx} and u_{yy}.

We are now left to derive a suitable scheme for u_{xy}. To do so, consider:

$$
\begin{aligned}
&u(x+\Delta x, y+\Delta y) - u(x-\Delta x, y+\Delta y) - u(x+\Delta x, y-\Delta y) + u(x-\Delta x, y-\Delta y) \\
&= 4\Delta x\Delta y\frac{\partial^2 u}{\partial x\partial y} + \mathrm{O}(\text{4th order derivatives error terms}).
\end{aligned}
$$

As such, we can write (the last two equations follow by exploiting the symmetry of the indices):

$$
u_{xy} = \frac{u(x+\Delta x, y+\Delta y) - u(x-\Delta x, y+\Delta y) - u(x+\Delta x, y-\Delta y) + u(x-\Delta x, y-\Delta y)}{4\Delta x\Delta y}, \quad (10)
$$

$$
u_{xz} = \frac{u(x+\Delta x, z+\Delta z) - u(x-\Delta x, z+\Delta z) - u(x+\Delta x, z-\Delta z) + u(x-\Delta x, z-\Delta z)}{4\Delta x\Delta z}, \quad (11)
$$

$$
u_{yz} = \frac{u(y+\Delta y, z+\Delta z) - u(y-\Delta y, z+\Delta z) - u(y+\Delta y, z-\Delta z) + u(y-\Delta y, z-\Delta z)}{4\Delta y\Delta z}. \quad (12)
$$

Additional Materials:

- Interactive tool: `https://kusumaatmaja.github.io/PDE3/examples`
- If you are interested, higher order stencil using more grid points to estimate the partial derivatives are possible. See for example:
  `https://en.wikipedia.org/wiki/Five-point_stencil` and
  `https://en.wikipedia.org/wiki/Nine-point_stencil`.

Version Control:
- version 1.0 on 20 December 2025